Robust LP: Discrete Failures - I

$$\min: \; c^T x$$

$$\text{st}: \; Ax \le b$$

$$a_R^T x \le b_R, \quad \forall a_R \in \underline{D_k}$$

$D_k = \{ a : a_i \in [\hat{a}_i - \delta_i, \hat{a}_i + \delta_i]$, and at least $n-k$ of the a_i's exactly equal $\hat{a}_i \}$

$\{\hat{a}_i, \delta_i, k\}$ all known.

Robust LP: Discrete Failures - II

$$\min: c^T x$$
$$\text{st: } Ax \le b$$

$$\begin{bmatrix} \max: a_R^T x \\ \text{st: } a_R \in D_R \end{bmatrix} \le b_R$$

$$\sum (\hat{a}_i + \delta_i z_i) x_i$$

$$\max: \sum a_i x_i \quad \Longleftrightarrow \quad \max: \sum \hat{a}_i x_i + \sum \delta_i z_i \cdot x_i$$

$$\text{st: } a_R \in D_R \qquad \text{st: } -1 \le z_i \le 1$$

at most R z_i's $\neq 0$

Robust LP: Discrete Failures - III

$$\max: \sum \hat{a}_i x_i + \sum \delta_i z_i x_i$$

$$\text{st: } -1 \le z_i \le 1$$

at most k z_i's $\neq 0$

$$\max: \sum \hat{a}_i x_i + \sum \delta_i |x_i| z_i$$

$$\text{st: } 0 \le z_i \le 1 \qquad 2n$$

~~at most k z_i's $\neq 0$~~

$$\sum z_i \le k \qquad 1$$

LP

n variables: $z_1, \ldots, z_n$

- Opt sol'n at ext pt
- Ext pts are defined by n ind constraints.

At least $n-1$ of the z_i's are integral: 0 or 1

but also: $\sum z_i = k \Rightarrow$ all z_i's are in fact integral

Robust LP: Discrete Failures - IV

Putting this all together:

Take the dual of LP on previous slide:

$$\left[\begin{array}{ll} \max: & \sum \delta_i |x_i| \cdot z_i + \sum \hat{a}_i x_i \\ \text{st:} & 0 \le z_i \le 1 \\ & \sum z_i \le k \end{array}\right] \le b_R$$

Take the dual

$$\begin{array}{ll} \min: & c^T x \\ \text{st:} & Ax \le b \end{array}$$

$$\left\{\begin{array}{l} p_0 \cdot k + \sum p_i \le b_R - \sum a_i x_i \\ p_0 + p_i \ge \delta_i \cdot |x_i| \\ p \ge 0 \end{array}\right.$$